Hongcheng Jiang

Applied AI Engineer | LLM Agents & Evaluation

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Kansas City, MO | Open to relocation / remote | U.S. Permanent Resident (no sponsorship required)

Summary

Applied AI Engineer with a Ph.D. in Electrical and Computer Engineering and experience spanning industry ML engineering and applied AI research. Builds LLM agent systems with LangGraph orchestration, hybrid retrieval, SQL verification, and durable Python services. Improved SQL correctness from 43.8% to 72.9% on 48 fixed development cases and reduced tokens per correct task by 35.9%, alongside research in medical measurement agents and geometry-conditioned visual models.

Technical Skills

Agents: LangGraph, MCP, Tool Calling, Hybrid RAG, Reranking, Bounded Repair, Structured Outputs, Evaluation Harnesses, Guardrails, Execution Tracing

Engineering: Python, SQL, FastAPI, SQLite, PostgreSQL, Async Workers, Docker, Linux, Git

ML: PyTorch, Hugging Face, QLoRA/PEFT, vLLM, Ollama, Vision Transformers, Diffusion Models

Professional Experience

University of Missouri–Kansas City — Jun 2026 – Present

Research Associate (Postdoc) — LLM Agents & Medical AI — *Kansas City, MO*

- RadMeasure — Medical Measurement Agent: Built a **KEEP/REPAIR/STOP controller** to constrain LLM actions to registered protocols and tools, verify measurement geometry, and route unresolved cases to human review; bounded repair attempts and recorded execution traces for audit.
- RadMeasure offline study: A learned repair policy selected 106/528 predictions (20.1%) across 176 archived cases; **mean error fell 0.69° among selected records**, with overall MAE decreasing from 2.68° to 2.54°.
- GeoVIRA (submitted, WACV 2027): Conditioned frozen visual features on geometry priors, cutting geometry-only MAE **13.5–25.1%** across three clinical angle tasks (hallux valgus, intermetatarsal, spinal Cobb); isolated the visual contribution with matched-versus-shuffled controls.

NextTier — IT Solutions Consultancy — Jun 2025 – May 2026

Machine Learning Engineer — *Sacramento, CA (Remote)*

- Built SQL-Agent, initially an internal text-to-SQL prototype; evolved it into a **LangGraph Planner–Verifier agent** with browser/FastAPI/MCP interfaces, bounded repair, user clarification, durable jobs, and human-approved database mutations.
- Implemented **BM25/vector hybrid retrieval + cross-encoder reranking** with database/table filtering. Improved retrieval Hit@1 from **68.8% to 93.8%** on 16 fixed queries.
- On 48 fixed development cases across six task families (BM25), improved SQL correctness **43.8%→72.9% (21/48→35/48)** and reduced tokens per correct task **35.9% (6,629→4,250)**, while client p95 latency remained approximately **5.9 s**.
- Implemented gateway-isolated approved writes, transactional idempotency, and durable async jobs; verified recovery under worker termination, restart, duplicate requests, and timeouts. Completed **100/100 scripted jobs at concurrency 8** in local Docker and passed **402 automated tests**.

University of Missouri–Kansas City — Jan 2020 – May 2025

Graduate Research Assistant — *Kansas City, MO*

- Developed transformer- and diffusion-based models for **thermal super-resolution, hyperspectral pansharpening, and NIR-to-RGB translation**; evaluated image quality and spectral fidelity.
- Co-authored a fully connected LSTM-CRF for clinical NER, reaching **84.15% F1 on i2b2/VA**.

Education

University of Missouri–Kansas City

Ph.D., Electrical & Computer Engineering, GPA: 4.0/4.0 — May 2025

M.S., Computer Science, GPA: 3.8/4.0 — Dec 2019

Publications

8 peer-reviewed publications; 6 first-author imaging papers. Full list: Google Scholar.